

A Two-Dimensional Counterexample to the Unrestricted Stability-Radius Representation

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Status. Candidate counterexample, likely valid. This packet gives a full negative answer to the explicit open problem in Badea–Mbekhta [1]; it is subject to human novelty and proof review.

1 Source problem

For bounded operators $T, S : X \rightarrow Y$, Badea and Mbekhta define the stability radius $d(T; S)$ using the punctured disk around zero. For Fredholm T it is the supremum of the radii $r > 0$ on which the nullity and deficiency of $T - \lambda S$ are constant for $0 < |\lambda| < r$.

Their Theorem 5.1 assumes that $\dim \mathcal{N}(T - \lambda S)$ is also constant at $\lambda = 0$, equivalently that the stability number $k(T; S)$ is zero, and proves

$$\begin{aligned} d(T; S) &= \sup \left\{ \frac{1}{r(SL)} : TLT = T \right\} \\ &= \sup \left\{ \frac{1}{r(SL)} : TLT = T, LTL = L \right\}. \end{aligned} \tag{1}$$

Here $r(\cdot)$ denotes spectral radius. In the introduction, page 2, and again in Remark 5.4, page 13, the paper asks whether (1) remains valid without the local-nullity constancy hypothesis.

It was proved in [BL] that if T is Fredholm, then the limit (1.2) exists and coincides with the so-called stability radius $d(T; S)$ of T and S . In the case when T is Fredholm and $\dim N(T - \lambda S)$ is constant in a neighborhood of $\lambda = 0$, the stability radius $d(T; S)$ equals the supremum of all $r > 0$ such that $\dim N(T - \lambda S)$ and $\operatorname{codim} R(T - \lambda S)$ are constant for $|\lambda| < r$.

A different representation formula was given recently [BM1] for the limit (1.1). It was proved there that if T is Fredholm and $0 \in \operatorname{reg}(T)$, the generalized resolvent set of T , then

$$\lim_{n \rightarrow \infty} \gamma(T^n)^{1/n} = \sup \left\{ \frac{1}{r(L)} : TLT = T \right\}, \quad (1.3)$$

where $r(L)$ is the spectral radius of L . The same result (1.3) is true without the condition of Fredholmness of T for Hilbert space operators [BM2].

The aim of the present note is to extend to linear pencils the formula (1.3). Namely, we will prove that

$$\begin{aligned} d(T; S) &= \sup \left\{ \frac{1}{r(SL)} : L \in \mathcal{L}(Y, X), TLT = T \right\} \\ &= \sup \left\{ \frac{1}{r(SL)} : L \in \mathcal{L}(Y, X), TLT = T, LTL = L \right\}, \end{aligned}$$

if T is Fredholm and $\dim N(T - \lambda S)$ is constant in a neighborhood of $\lambda = 0$. The above formula relates the limit (1.2) with the spectral radii of generalized inverses of T . It is an open problem if this representation holds without the condition of constancy of $\dim N(T - \lambda S)$.

One of the ingredients of the proof of the main result is the existence of generalized resolvents for some operator pencils. This is interesting in its own.

The paper is organized as follows. In the next section we recall

Figure 1: Source page 2: the representation and the question about dropping the constancy hypothesis.

2 Idea of the proof

The omitted hypothesis permits the pencil to gain rank immediately away from zero. A diagonal 2×2 pencil makes this jump transparent and places its next singular point at distance one, so its stability radius is finite. Meanwhile, the singular operator T has a one-parameter family of generalized inverses whose nonzero eigenvalue tends to zero. Their spectral radii therefore force both right-hand sides of (1) to be infinite.

3 Counterexample

Theorem 1. *Let $X = Y = \mathbb{C}^2$, equipped with any norm, and set*

$$T = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad S = I_2.$$

Then T is Fredholm, $k(T; S) = 1$, and $d(T; S) = 1$. Nevertheless,

$$\sup_{TLT=T} \frac{1}{r(SL)} = \sup_{\substack{TLT=T \\ LTL=L}} \frac{1}{r(SL)} = +\infty.$$

Consequently, the constancy hypothesis in the source theorem cannot be removed.

Proof. Every operator on the finite-dimensional space $\mathbb{C}^2$ is bounded, and T is Fredholm. The pencil is

$$T - \lambda S = \begin{pmatrix} 1 - \lambda & 0 \\ 0 & -\lambda \end{pmatrix}, \quad \det(T - \lambda S) = \lambda(\lambda - 1).$$

Thus $T - \lambda S$ is invertible for every $\lambda \notin \{0, 1\}$. At $\lambda = 1$ it has nullity and deficiency one. Hence, on every punctured disk $0 < |\lambda| < r$ with $r \leq 1$, the nullity and deficiency are both zero, whereas every disk with $r > 1$ contains the exceptional point $\lambda = 1$. By the source definition,

$$d(T; S) = 1.$$

The omitted assumption genuinely fails. Indeed, $\mathcal{N}(T) = \text{span}\{e_2\}$, while $\dim \mathcal{N}(T - \lambda I_2) = 0$ for all sufficiently small nonzero λ . Directly from the source definition of $k(T; S)$, the decreasing spaces satisfy

$$R_1 = S^{-1}TR_0 = \mathcal{R}(T) = \text{span}\{e_1\}, \quad R_m = \text{span}\{e_1\} \quad (m \geq 1).$$

Therefore $X_\infty = \text{span}\{e_1\}$ and

$$k(T; S) = \dim \frac{\mathcal{N}(T)}{\mathcal{N}(T) \cap X_\infty} = 1.$$

Now fix $\varepsilon > 0$ and put

$$L_\varepsilon = \begin{pmatrix} 1 & 1 \\ -1 + \varepsilon & -1 + \varepsilon \end{pmatrix}.$$

Straight multiplication gives

$$TL_\varepsilon T = T, \quad L_\varepsilon TL_\varepsilon = L_\varepsilon.$$

Thus L_ε is admissible in both suprema in (1). A second multiplication gives

$$L_\varepsilon^2 = \varepsilon L_\varepsilon.$$

Equivalently, $\det L_\varepsilon = 0$ and $\operatorname{tr} L_\varepsilon = \varepsilon$, so its eigenvalues are 0 and ε . Since $S = I_2$,

$$r(SL_\varepsilon) = r(L_\varepsilon) = \varepsilon.$$

Each right-hand supremum is therefore at least $1/\varepsilon$ for every $\varepsilon > 0$, hence is $+\infty$. This contradicts the value $d(T; S) = 1$ and proves the claim. $\square$

4 Verification and robustness

The script `code/verify_matrices.py` checks all displayed matrix identities over the rational numbers for several positive values of ε . It also checks $\det(T - \lambda I_2) = \lambda(\lambda - 1)$.

At $\varepsilon = 0$ one obtains the nilpotent generalized inverse

$$L_0 = \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}, \quad L_0^2 = 0.$$

The positive- ε family is included to avoid any dependence on a convention for $1/0$: all terms $1/r(SL_\varepsilon)$ are finite, but they are unbounded.

The example works with any norm on $\mathbb{C}^2$, so it is simultaneously a Banach-space and Hilbert-space counterexample.

5 Scope and novelty check

This result only disproves the proposed removal of the condition $k(T; S) = 0$. It does not affect Theorem 5.1 under its stated hypotheses and does not exclude a modified formula with extra restrictions on L .

Before promotion, the run's registry, solution, attempt, and proof-gap indexes and its parsed arXiv source corpus were searched for arXiv:math/0011229, the exact paper title, the displayed $r(SL)$ formula, and the constancy condition. Bounded web searches used the exact title, DOI 10.1006/jmaa.2000.7445, open-problem wording, and formula fragments. They returned the source and later work on generalized resolvents citing it, but no claimed resolution or matching counterexample. The search was not an exhaustive citation review, so the novelty assessment is provisional.

6 Human-review recommendation

Verify the source's extended-real convention and punctured-disk definition, then check the four 2×2 identities in Theorem 1. The $\varepsilon > 0$ family makes the conclusion independent of the former convention. If those checks pass, this packet gives a full negative answer to the source open problem.

References

- [1] C. Badea and M. Mbekhta, *The stability radius of Fredholm linear pencils*, J. Math. Anal. Appl. 260 (2001), 159–172; arXiv:math/0011229v2.